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Math 544/CptS. 531 - Fall 2024

Take home midterm exam

Work on your own. You may consult textbooks and lecture notes. To receive full credit, explain your steps and display your computations explicitly, within reason. You may use Matlab for Problem 1 (c). Cite any help you obtained from a chatbot.

1. Let $A = \begin{bmatrix} 2 & -1 & 1 \\ 6 & 1 & 2 \\ 2 & -1 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$. **[30 pts]**

   (a) Compute a lower triangular matrix L and upper triangular matrix U such that $A = LU$.

   (b) Make use of L and U to solve the system $Ax = b$.

   (c) Consider the iterative Jacobi method to solve the same system. Does it work? If not, think of preconditioning the system so that the Jacobi method works and perform 4 steps of the iteration. (Hint: Consider a permutation of the rows of A.)

2. Find a least-squares solution of the system $Ax = b$, where $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}$, $b = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$. Use any method you like. **[30 pts]**

3. Consider a subspace W of $\mathbb{R}^n$. **[40 pts]**

   (a) Give a precise definition of the orthogonal complement, $W^\perp$, of W.

   (b) Prove that $W^\perp$ is a subspace of $\mathbb{R}^n$.

   (c) Find an orthonormal basis of $W = \text{span}\{x, y\}$, where $x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$.

   (d) Find the orthogonal projection of $z = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ onto the subspace W in (c).

Solution:

Problem 1.

Let

(1) $$A = \begin{bmatrix} 2 & -1 & 1 \\ 6 & 1 & 2 \\ 2 & -1 & 3 \end{bmatrix} \quad \&$$

(2) $$b = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}.$$

(a) Finding L & U s.t $A = LU$:

let me start row ops on $[A \mid b]$. We want to do row ops on the augmented matrix so that to check our result at the end:

$$e_{ip} = -\frac{r_{ip}}{r_{pp}}$$

← row i operation entry

→ pivot row entry

$$[A \mid b] = \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 6 & 1 & 2 & 2 \\ 2 & -1 & 3 & 0 \end{array}\right] \xleftarrow[e_{21} = -\frac{6}{2} = -3]{R2 = r2 - 3r1} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 0 & 4 & -1 & -4 \\ 2 & -1 & 3 & 0 \end{array}\right]; \quad E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(3) $$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 0 & 4 & -1 & -4 \\ 2 & -1 & 3 & 0 \end{array}\right] \xleftarrow[e_{31} = -\frac{2}{2} = -1]{R3 = r3 - r1} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 0 & 4 & -1 & -4 \\ 0 & 0 & 2 & -2 \end{array}\right]; \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\underbrace{}_{\text{already } U} \qquad \underbrace{}_{c}$$

So we've come to the upper triangular matrix U with

$$E_{31} \cdot E_{21} [A \mid b] = [U \mid c] \quad \text{where}$$

(4) $$U = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 2 \end{bmatrix} \quad \& \quad c = \begin{bmatrix} 2 \\ -4 \\ -2 \end{bmatrix}.$$

To obtain L we need:

(5) $$E_{31} \cdot E_{21} \cdot A = U \quad \text{where}$$

$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ & $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Since these E_{ij} are elementary matrices, they <u>do have</u> inverses E_{ij}^{-1} that undo what E_{ij}'s do. So they're invertible. Then

$$E_{31} \cdot E_{21} \cdot A = U \quad \Big\} * E_{31}^{-1} \text{ by left} \Rightarrow$$

$$\underbrace{(E_{31}^{-1} \cdot E_{31})}_{I} \cdot E_{21} \cdot A = E_{31}^{-1} \cdot U \quad \Rightarrow$$

$$E_{21} \cdot A = E_{31}^{-1} \cdot U \quad \Big\} * E_{21}^{-1} \text{ by left} \Rightarrow$$

(6) $$\boxed{A = E_{21}^{-1} \cdot E_{31}^{-1} \cdot U}$$

To calculat (6):

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}^{-1} \cdot U = \overbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}^{\text{Note: } E_{31}^{-1}} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \cdot U \Rightarrow$$

$$\Rightarrow A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}}_{\text{let } L} \cdot U := L \cdot U, \quad \text{so}$$

(7) $$\boxed{A = L \cdot U}, \quad \text{where } L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \text{ \& } U = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

(8) Use L & U found in (7) for solving (1):

$$\underbrace{Ax}_{\downarrow} = b \quad \text{can now be written}$$

$$\overbrace{LU}\,x = b \quad \Big\{ \cdot \overbrace{L^{-1}}^{\text{exist because } L \text{ is a product of invertible matrices}} \text{ of BS by left:}$$

$$L^{-1} \cdot L\,U\,x = L^{-1} \cdot b$$

(5) $$U\,x = L^{-1} b \quad \text{where by (2) \& (7)}$$

$$\underbrace{\begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 2 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}}_{L^{-1}} \cdot \underbrace{\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}}_{b} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

(6) $$\begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ -2 \end{bmatrix} \Rightarrow$$

← matching the c in (4) ✓

Solving this (6) by back substitution:

$$2x_3 = -2 \quad \Rightarrow \quad x_3 = -1$$
$$4x_2 - x_3 = -4 \quad \Rightarrow \quad x_2 = -5/4$$
$$2x_1 - x_2 + x_3 = 2 \quad \Rightarrow \quad x_1 = 7/8$$

(7) $$[x_1 \;\; x_2 \;\; x_3]^T = \left[\frac{7}{8}, \; \frac{-5}{4}, \; -1\right]^T$$

checking:

$$\begin{bmatrix} 2 & -1 & 1 \\ 6 & 1 & 2 \\ 2 & -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 7/8 \\ -5/4 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{7}{4} + \frac{5}{4} - 1 \\ \frac{21}{4} - \frac{5}{4} - 2 \\ \frac{7}{4} + \frac{5}{4} - 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} \equiv b \quad \text{in (2). ✓.}$$

MATH 544. MIDTERM

Problem 1.

Let A be the matrix defined below.

- (a) Compute a lower triangular matrix L and upper triangular matrix U such that A = LU.
- (b) Make use of L and U to solve the system Ax = b.
- (c) Consider the iterative Jacobi method to solve the same system. Does it work? If not,think of preconditioning the system so that the Jacobi method works and perform 4 steps of the iteration. (Hint: Consider a permutation of the rows of A.)

Ax = b where

```
[ 2 -1  1][x1]   [ 2]

[ 6  1  2][x2] = [ 2]

[ 2 -1  3][x3]   [ 0]
```

Solution:

```
# To solve Ax=b, We start by constructing the arrays for A and b.
import numpy as np
from numpy.linalg import eig

A = np.array([[ 2, -1,  1],
              [ 6,  1,  2],
              [ 2, -1,  3]])
b = np.transpose(np.array([ 2,  2,  0]))
# To solve the system we do

x = np.linalg.solve(A,b)
print("Real solution: ")
print(x)
```

```
Real solution:
[ 0.875 -1.25  -1.   ]
```

Using Jacobi Iteration:

```
A = np.array([[ 2, -1,  1],
              [ 6,  1,  2],
              [ 2, -1,  3]])

L = np.array([[ 0,  0,  0],
              [ 6,  0,  0],
              [ 2, -1,  0]])

D = np.array([[ 2,  0,  0],
              [ 0,  1,  0],
              [ 0,  0,  3]])

U = np.array([[ 0, -1,  1],
              [ 0,  0,  2],
              [ 0,  0,  0]])

D_inverse =  np. linalg. inv(D)
b = np.transpose(np.array([ 2,  2,  0]))
D_inverse_b = np.dot(D_inverse, b)

BJ = np.dot(-D_inverse, L+U)
w,v=eig(BJ)
```

```
print('BJ evalues:', w)

x0 = np.transpose(np.array([ 0,  0,  0]))
x = list();
x.append(x0)

for i in range(10):
  x1 = np.dot(BJ, x0)+ D_inverse_b
  x0 = x1
  x.append(x1)
Aproximations = np.array(x)
print(Aproximations)
```

```
BJ evalues: [-0.23452149+1.87038347j -0.23452149-1.87038347j  0.46904298+0.j        ]
[[   0.            0.            0.        ]
 [   1.            2.            0.        ]
 [   2.           -4.            0.        ]
 [  -1.          -10.           -2.66666667]
 [  -2.66666667   13.33333333   -2.66666667]
 [   9.           23.33333333    6.22222222]
 [   9.55555556  -64.44444444    1.77777778]
 [ -32.11111111  -58.88888889  -27.85185185]
 [ -14.51851852  250.37037037    1.77777778]
 [ 125.2962963    85.55555556   93.13580247]
 [  -2.79012346 -936.04938272  -55.01234568]]
```

The Jacobi Iteration applied to the given matrix A did not converge, as indicated by the solution vector [−2.7901,−936.0494,−55.0123], which shows magnitudes that are blowing up. This non-convergence occurs because the matrix A is not well-preconditioned; specifically, partial pivoting has not been applied, meaning the largest absolute value in each column is not in the pivotal position.

To improve convergence, we will use partial pivoting by swapping rows 1 and 2 of A. This step will bring the largest absolute value in the first column to the top-left position, resulting in a preconditioned matrix that is better suited for iterative methods like the Jacobi Iteration.

After performing this row swap, we will proceed with the Jacobi Iteration on the new matrix, expecting it to yield a convergent solution due to the improved conditioning.

```
A = np.array([[ 6,  1,  2],
              [ 2, -1,  1],
              [ 2, -1,  3]])

# P = np.array([[ 0,  1,  0],
#               [ 1,  0,  0],
#               [ 0,  0,  1]])

# A = np.dot(P, A)

L = np.array([[ 0,  0,  0],
              [ 2,  0,  0],
              [ 2, -1,  0]])

D = np.array([[ 6,  0,  0],
              [ 0, -1,  0],
              [ 0,  0,  3]])

U = np.array([[ 0,  1,  2],
              [ 0,  0,  1],
              [ 0,  0,  0]])

D_inverse =  np. linalg. inv(D)
b = np.transpose(np.array([ 2,  2,  0]))
# b = np.dot(P, b)
D_inverse_b = np.dot(D_inverse, b)

BJ = np.dot(-D_inverse, L+U)
w,v=eig(BJ)

print('BJ evalues:', w)

x0 = np.transpose(np.array([ 0,  0,  0]))
x = list();
x.append(x0)

for i in range(10):
  x1 = np.dot(BJ, x0)+ D_inverse_b
  x0 = x1
```

```
  x.append(x1)
Aproximations = np.array(x)
print(Aproximations)
```

```
BJ evalues: [-0.6310964+0.j          0.3155482+0.27656785j  0.3155482-0.27656785j]
[[ 0.          0.          0.        ]
 [ 0.33333333 -2.          0.        ]
 [ 0.66666667 -1.33333333 -0.88888889]
 [ 0.85185185 -1.55555556 -0.88888889]
 [ 0.88888889 -1.18518519 -1.08641975]
 [ 0.89300412 -1.30864198 -0.98765432]
 [ 0.88065844 -1.20164609 -1.03155007]
 [ 0.8774577  -1.2702332  -0.98765432]
 [ 0.87425697 -1.23273891 -1.00838287]
 [ 0.87491744 -1.25986892 -0.99375095]
 [ 0.8745618  -1.24391607 -1.0032346 ]]
```

As expected, the approximated soltuion [0.8745618 -1.24391607 -1.0032346] is approaching to the real solution [0.875 -1.25 -1.] that we've found earlier.

```
# !sudo apt-get install texlive-xetex texlive-fonts-recommended texlive-plain-generic
```

```
# !jupyter nbconvert --to pdf /content/Math544_Midterm_Lazizbek.ipynb
```

Start coding or generate with AI.

Problem 2. Find a least-squares sol-n of the system $Ax = b$, where

(1) $$A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}.$$

Use any method.

Sol-n:

(i) First easy way to do this Problem might be by normal equations, A^TA; given

$$Ax = b \quad \text{f. } A^T \text{ by left}$$

(2) $$A^T \cdot A\hat{x} = A^T b$$

Claim is the solution $\hat{x}$ for normal equation in (2) is the least-squares solution for (1) as $R(A^TA) \equiv R(A)$.

Thus

(3) $$A^TA = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}^T \cdot \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix} \quad \&$$

$$A^Tb = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}^T \cdot \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Then

$$A^TA\hat{x} = A^Tb \quad \text{becomes}$$

(4) $$\begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \Rightarrow$$

(5) $$\hat{x} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Problem 3.

Consider a <u>subspace W</u> of $\mathbb{R}^n$.

(a) Give a precise definition of the orthogonal complement, $W^\perp$ of W.

Sol-n:

Let W be a subspace of $\mathbb{R}^n$.

<u>Definition</u> ($W^\perp$, orthogonal complement of W):

The orthogonal complement, $W^\perp$ of a subspace W is

(1) $$S^\perp = \{ x \in \mathbb{R}^n : \langle x, y \rangle = 0 \ , \forall y \in W \}.$$

(b) Prove that $W^\perp$ is a subspace of $\mathbb{R}^n$. $\boxtimes$

Based on 3 properties of a subspace, let me verify $W^\perp$ meets them:

(i) $0 \in W^\perp$ because

$x \in W^\perp$, $\langle x, 0 \rangle = 0$, $0 \in W$ since it's a subspace.

(ii) Let $x_1, x_2 \in W^\perp$. Then by its defined way in (1)

(2) $\langle x_1, y \rangle = 0 \quad \forall y \in W$ &

(3) $\langle x_2 \ y \rangle = 0 \quad \forall y \in W$.

Now consider

$$\langle x_1 + x_2, y \rangle = \underbrace{\langle x_1, y \rangle}_{=0 \text{ by } (2)} + \underbrace{\langle x_2, y \rangle}_{=0 \text{ by } (3)} = 0 + 0 = 0 \Rightarrow$$

(4) $x_1 + x_2 \in W^\perp$.

(iii) Finally, let $x \in W^{\perp}$. $\Rightarrow$

(5) $\langle x, y \rangle = 0$, $\forall y \in W$.

And consider $c \in \mathbb{R}$ with

$\langle cx, y \rangle = c\langle x, y \rangle = c \cdot 0 = 0$, $\Rightarrow$

(6) $cx \in W^{\perp}$,

thus by (i), (ii), (iii), $W^{\perp}$ is a subspace of $\mathbb{R}^n$. $\boxtimes$

(c) Find an orthonormal basis of W,

$$W = \text{span}\{x, y\} \text{ wher } x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \ \& \ y = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

Sol-n. By Gram-Schmidt process, let me try to find the orthonormal basis $\{u_1, u_2\}$ s.t.

(7) $$W = \text{span}\{x, y\} = \text{span}\{u_1, u_2\}.$$

$$v_1 = x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \ \&$$

$$v_2 = y - \frac{\langle y, v_1 \rangle}{\langle v_1, v_1 \rangle} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} - \frac{\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} - \frac{2}{2} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

So we have orthogonal set $\{v_1, v_2\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$. Let's normalize it

(8) $$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \ \& \ u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Thus for (7), we have

$$W = \text{span}\{x, y\} = \text{span}\{u_1, u_2\} = \text{span}\left\{ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}. \ \boxtimes$$

(d) Find the orthogonal projection of

(9) $$z = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

onto the subspace W in (c).

Sol-n: By the Orthogonal Decomposition Theorem (ODT) below, we can find $\text{proj}_W(z)$.

Theorem (ODT):
Let $W \subseteq \mathbb{R}^n$ be a subspace. Then, $\forall x \in \mathbb{R}^n$, $\exists$ unique $w \in W$, $\hat{w} \in W^\perp$ such that $x = w + \hat{w}$.
Moreover,

(10) $$w = \sum_{i=1}^{k} \langle x, u_i \rangle u_i$$

where $\{u_1, u_2, .. u_k\}$ is an orthonormal basis for W, & $\hat{w} = x - w$,
(w = "orthogonal projection of x onto W, i.e. $w = \text{proj}_W(x)$.).

Then by this Thm (ODT) & our obtained orthonormal basis in (8),

$$w = \text{proj}_W(z) = \sum_{i=1}^{2} \langle z, u_i \rangle \cdot u_i = \langle z, u_1 \rangle u_1 + \langle z, u_2 \rangle u_2 =$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \sqrt{2} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} + 0 \cdot \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

(11) $$w = \text{proj}_W(z) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

$\boxtimes$